

Prospectus

Jonah Cooke

Advisor: Yuan Shi

Plasma Quantum Group

Department of Physics, University of Colorado at Boulder

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Brief background to dark matter

In 1980, Vera Rubin published a paper that is credited with providing the first substantial evidence for dark matter (Rubin, Ford, and Thonnard 1980). However, no one has been able to identify the unknown substance beyond speculation. Reasonable efforts have failed to describe the rotation curves Rubin first noticed, which implies that a new direction needs to be tried. Modern plasma physics gives insight on what to try.

Background to research

When a photon travels through a plasma medium, its energy is conserved. This means that the plasma frequency and wave number of the photon must change inversely, where the plasma frequency is dependent on the density of the plasma. This behavior means that as a photon travels through a plasma with varying density, if the plasma density goes up, the wave number of the photon must go down. Using Snell's law, the photon's angle must increase away from the vertical, meaning that through a plasma gradient it diverges from its path of travel. This behavior is based on the equation:

$$\omega^2 = \omega_p^2 + k^2$$

(natural units assumed). This is very similar in form to Einstein's equation:

$$E^2 = m^2 + p^2.$$

This implies that we may be able to treat mass as a variable and obtain a similar effect. However, the treatment is more complex, as the momentum is inherently mass dependent and the opposite dependence is necessary for the observed galactic curves.

My planned research

A detailed treatment of the one-dimensional case can be found in (Shi 2021). However, the interaction of orbital bodies is two-dimensional, and work needs

to be done to extend the research to two dimensions. This can be done by modifying the Schwarzschild metric to encompass a non-constant mass distribution.

The one-dimensional case gave a differential equation for the mass distribution, and using a known trajectory I was able to find a mass distribution that agreed with that trajectory. The two-dimensional case is inherently more difficult. Although an analytic solution exists (Hackmann and Lämmerzahl 2008), it is much more common to use numerical solutions because of the complexity of the system. In addition to having the potential to describe the observed behavior of sparse matter in galaxies, this approach offers greater flexibility and is guaranteed, by the structure of the method, to yield an analytic solution. It is likely to be as complex as, or more complex than, a Schwarzschild geodesic, but it retains the generality offered by an analytic solution. An analytic solution will allow for a general comparison to observed data and theoretical results, meaning that if uniqueness can be proven for the result, it will either be a solution to all observed trajectories or to only one observed trajectory.

Schedule

By the end of May, I plan to have the theoretical framework complete (that is, to have a “geodesic equation”), and during June I plan to begin comparing the results to other trajectories based on their initial conditions. I will write most of my thesis during July.

Bibliography

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